

Arc length of curves

Key Points

The arc length for a function $y(x)$ is

$$\int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

With limits in x

The arc length for parametric curves $x(t), y(t)$ is

$$\int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

With limits in t

Basic Skills

Q1. Explain how to work out the distance between points

Q2. For the function $y = 2x$

a) Find the distance between points where $x = 0, 5$

b) Show that the integral for the distance can be written as $\int_0^5 \sqrt{5} dx$ and hence show the distance is the same as part a)

Q3. For the parametric equation $x(t) = t^2, y(t) = t^3$

a) Show that the integral $\int_0^1 \sqrt{4t^2 + 9t^4} dt$ calculates the arc length for times 0 to 1

b) Hence calculate the arc length between points (0,0), (1,1)

Competency Skills

Q4. The parabola given by $x(t) = 2t^2 + 1, y(t) = 4t$ for $0 \leq t \leq 2$

a) By using the parametric show that the length of the parabola is given by $\int_0^2 4\sqrt{t^2 + 1} dt$

b) Hence by using the substitution $t = \sinh u$ find the length of the parabola

Q5. Find the length of the hyperbolic cosine functions between $x = 0, \ln a$ in terms of a

Q6. The asteroid is given by $x(\theta) = a \cos^3 \theta, y(\theta) = a \sin^3 \theta$ for $0 \leq \theta \leq 2\pi$ for some a

a) Sketch the asteroid

b) Calculate the length of the asteroid in terms of a

Q7. For $x \in [0, k]$ the length of the exponential growth curve is 1, find an equation for k

Advanced Skills

Q8. A cycloid is created by tracking the height of a fixed point on a circle of radius r as it rolls

a) Given the circle is of radius r and performs a full rotation every 2π units of time explain why the parametric form of the cycloid can be written as $x = r(t - \sin t), y = r(1 - \cos t)$

b) Prove that in one rotation of the circle the length of the cycloid is $8r$

Q9. By using the parametric form of an ellipse $x = a \cos \theta, y = b \sin \theta$, with eccentricity $e = \frac{\sqrt{a^2 - b^2}}{a}$ for $0 \leq \theta \leq 2\pi$

a) Show that the perimeter is given by $P = 4a \int_0^{\frac{\pi}{2}} \sqrt{1 - e^2 \cos^2 \theta} d\theta$

b) Hence show for small eccentricities $P \approx 2\pi a \left(1 - \frac{1}{4}e^2\right)$

Extension

Q10. Show that the perimeter of an ellipse is equivalent to the length of one period of $k \cos(cx)$ determining the constants k, c in terms of a, b